Internet of Things (IoT) with Applications

Undergraduate Engineering Course — Syllabus, Lecture Plan and Laboratory Manual

Course Objective

To introduce students to IoT concepts, architectures, communication technologies, sensors, edge/cloud computing, IoT security, and practical applications in smart agriculture, healthcare, industry, smart cities, energy, transportation and education.

UNIT I: Introduction to IoT

Introduction to IoT: Definition and characteristics of IoT Evolution of IoT , IoT vs Internet , IoT vs M2M , IoT vs Cyber-Physical Systems , Components of IoT , IoT enabling technologies

IoT Architecture: Three-layer architecture , Five-layer architecture , Perception layer , Network layer , Processing/Cloud layer , Application layer , Edge and Fog computing

IoT Devices: Sensors , Actuators , Microcontrollers , Embedded systems , RFID , NFC , GPS , Smart devices

IoT Data: Data acquisition , Data transmission , Data storage , Data analytics , Real-time monitoring , IoT dashboards

UNIT II: IoT Communication and Networking

Communication Technologies – Short Range: Bluetooth/BLE , Zigbee , RFID , NFC , Wi-Fi

Communication Technologies – Long Range: LoRaWAN , NB-IoT , LTE-M , 4G/5G

IoT Protocols: MQTT , CoAP , HTTP/HTTPS , AMQP , WebSockets

IoT Network Concepts: IPv4 and IPv6 , 6LoWPAN , Wireless Sensor Networks , Gateway , Addressing and routing

UNIT III: IoT Platforms, Edge and Cloud

IoT Hardware Platforms: Arduino , ESP32 , Raspberry Pi , NodeMCU , Sensors and modules

Cloud Computing for IoT: IoT cloud architecture , Device management , Data storage , Cloud analytics , IoT dashboards

Edge Computing: Need for edge computing , Edge vs Cloud , Real-time processing , Edge AI , Fog computing

IoT + AI: Machine learning with sensor data , Anomaly detection , Predictive maintenance , Computer vision , Intelligent IoT

Example pipeline: Sensor → ESP32 → Edge AI → Cloud → Dashboard → Actuator

UNIT IV: IoT Security and Data Analytics

IoT Security: Security requirements , Authentication , Authorization , Encryption , Secure communication , Device security , Network security , Cloud security

IoT Threats: Device attacks , Eavesdropping , Spoofing , Denial-of-Service attacks , Malware , Data leakage

IoT Data Analytics: Data preprocessing , Time-series data , Visualization , Anomaly detection , Predictive analytics , AI/ML-based IoT analytics

Blockchain + IoT: Decentralized IoT , Smart contracts , Data integrity , IoT identity management

UNIT V: IoT Applications

1. Smart Agriculture

Sensors → IoT Gateway → Cloud → AI → Irrigation

Applications: Soil moisture monitoring , Automatic irrigation , Crop monitoring , Weather monitoring , Disease detection , Smart fertilizer management

2. Healthcare IoT

Applications: Remote patient monitoring , ECG monitoring , SpO_2 monitoring , Temperature monitoring , Smart medical devices , Elderly-care systems , Emergency alerts

3. Industrial IoT (IIoT)

Applications: Machine monitoring , Predictive maintenance , Asset tracking , Industrial automation , Digital twins , Quality monitoring

4. Smart City

Applications: Smart traffic management , Smart parking , Smart streetlights , Smart waste management , Air-quality monitoring , Water management

5. Smart Energy

Applications: Smart meters , Smart grid , Energy monitoring , Demand prediction , Renewable-energy monitoring , Building energy management

6. Smart Transportation

Applications: Vehicle tracking , Fleet management , Intelligent traffic systems , Connected vehicles , Accident detection , Predictive maintenance

7. Smart Home

Applications: Smart lighting , Smart security , Smart appliances , Energy management , Voice-controlled devices , Intrusion detection

30-Lecture Teaching Plan

#	Topic	#	Topic
1	Introduction to IoT	2	Characteristics and evolution
3	IoT architecture	4	Sensors and actuators
5	IoT devices	6	RFID, NFC and GPS
7	Wireless Sensor Networks	8	Wi-Fi, Bluetooth and Zigbee
9	LoRaWAN	10	NB-IoT and LTE-M
11	5G and IoT	12	MQTT
13	CoAP and HTTP	14	IoT gateways
15	Arduino and ESP32	16	Raspberry Pi
17	IoT cloud computing	18	Edge computing
19	Fog computing	20	IoT data analytics
21	AI + IoT	22	IoT security
23	Blockchain + IoT	24	Smart Home
25	Smart Agriculture	26	Healthcare IoT
27	Industrial IoT	28	Smart City
29	Smart Energy & Transportation	30	Case studies and mini-projects

IoT Laboratory — 15 Experiments

No.	Experiment	Description
1	Introduction to Arduino/ESP32	Read digital and analog inputs and control an LED.
2	Temperature and Humidity Monitoring	Use DHT11/DHT22 sensor and display readings.
3	Light Intensity Monitoring	Use an LDR to measure environmental light.
4	Ultrasonic Distance Measurement	Measure distance using HC-SR04.
5	Motion Detection	Use a PIR sensor to detect human movement.
6	IoT-Based Temperature Monitoring	Send sensor data through Wi-Fi to an IoT dashboard.
7	MQTT-Based IoT Communication	Publish sensor data to an MQTT broker and subscribe to it.
8	Smart Home Automation	Control lights/appliances remotely using ESP32.
9	IoT-Based Smart Agriculture	Measure soil moisture and automatically control a water pump.
10	IoT-Based Air Quality Monitoring	Measure air-quality parameters and display them on a dashboard.
11	IoT Healthcare Monitoring	Monitor temperature, pulse/SpO_2 and generate alerts.
12	GPS-Based Vehicle Tracking	Collect GPS coordinates and display vehicle location.
13	IoT Data Visualization	Send sensor data to a cloud platform and create graphs.
14	AI-Based IoT Anomaly Detection	Use Python/ML to identify abnormal sensor readings.
15	Mini Project	Develop an integrated IoT application (Smart Agriculture, Healthcare, Campus, Parking, Energy, or Industrial Predictive Maintenance).

Suggested Mini Projects

1. Smart Campus: *IoT Sensors → ESP32 → Wi-Fi → Cloud → AI Dashboard* — Monitor: Classroom occupancy, Energy consumption, Temperature, Air quality, Smart lighting

2. Smart Agriculture: *Soil Sensors → ESP32 → Cloud → AI → Automated Irrigation*

3. Smart Healthcare: *Wearable Sensors → Edge Device → Cloud → AI → Doctor Dashboard*

4. Smart Energy: *Smart Meter → IoT Gateway → Cloud → Energy Analytics → Alerts*

5. Industrial Predictive Maintenance: *Machine Sensors → Edge Device → AI Model → Failure Prediction*

IoT + AI + Edge + 5G: Future Direction

A useful way to conclude the course is to frame IoT's evolution in four stages: IoT 1.0 → Connected Things , IoT 2.0 → Cloud-connected Things , IoT 3.0 → AI-enabled IoT , IoT 4.0 → Edge AI + 5G + Digital Twins + Autonomous Systems.

Recommended Course Outcomes

After completing the course, students should be able to:

Code	Outcome
CO1	Explain IoT architecture, components and enabling technologies.
CO2	Select appropriate sensors, communication protocols and IoT platforms.
CO3	Develop IoT applications using Arduino/ESP32/Raspberry Pi.
CO4	Apply cloud, edge computing and AI techniques to IoT data.
CO5	Design secure IoT solutions for real-world applications.
CO6	Develop an application-oriented IoT mini-project.

This structure can be directly developed into a full 5-unit university syllabus, a 15-experiment IoT laboratory manual, PPT lecture notes, a question bank, and a 250–300 page IoT textbook.